

4.12.6. Reliable Message State

4.12.6.1. Retransmission Table

For retransmissions, the sender maintains a *retransmission table* of context records containing information on all reliable messages sent that have acknowledgments still pending. Each such reliable message context record includes the following fields:

- Reference to Exchange Context
- Message Counter
- Reference to fully formed, encoded and encrypted message buffer
- Send count
- Retransmission timeout counter

Each time a message that requires acknowledgment is sent, a new retransmission context record is inserted into the *retransmission table* or an existing record is updated to increment its send count. The message is sent a configurable maximum number of times (*MRP_MAX_TRANSMISSIONS*) and, if still undelivered, the application is notified of the failure.

4.12.6.2. Acknowledgement Table

The receiver maintains an *acknowledgement table* of context records containing information on each reliable message for which an acknowledgment SHALL be sent. Each such reliable message context record includes the following fields:

- Reference to Exchange Context
- Message Counter
- A boolean, **StandaloneAckSent**, indicating whether a **standalone acknowledgement** has been sent for this message counter. Initially false.

An entry SHALL remain in the table until one of the following things happens:

1. The exchange associated with the entry is closed. See [Section 4.10.5.3, “Closing an Exchange”](#).
2. The exchange associated with the entry has switched to track a pending acknowledgement for a new message counter value. See [Section 4.12.5.2.2, “Standalone acknowledgement processing”](#).
3. A message that is not a **standalone acknowledgement** is sent which serves as an acknowledgment for the entry. See [Section 4.12.5.1.1, “Piggyback acknowledgment processing”](#).

4.12.7. MRP Messages

4.12.7.1. MRP Standalone Acknowledgement

The *MRP Standalone Acknowledgement* message SHALL be formed as follows:

- The application payload SHALL be empty.
- The **A Flag** SHALL be set to **1**.